

A MATHEMATICAL MODEL OF DIVORCE IN MARRIAGE

**A PROJECT SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENT FOR THE AWARD OF THE BACHELOR OF
SCIENCE (B.Sc) DEGREE IN MATHEMATICS**

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NSUKKA**

BY

UBENYI CHINEMEREM PEACE

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SUPERVISOR: PROFESSOR G.C.E MBAH

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DECLARATION

I, Ubenyi Chinemerem Peace with registration number 2018/243520, hereby declare that this undergraduate project on "A mathematical modeling of divorce in marriage" for the degree, Bachelor of Science in the University of Nigeria contains my independent work and has not been previously submitted in any institution of higher learning to the best part of my knowledge.



Ubenyi Chinemerem Peace

2018/243520

DEDICATION

This work is dedicated to the almighty God for the wisdom and strength to carry out this work and to my wonderful parents; Mr and Mrs Cyprian Ubenyi for their support and prayers.

ACKNOWLEDGEMENT

I am forever grateful to the Almighty God who has shown me mercy and helped me throughout every stage of my work and also provided the right people around me.

I would not fail to acknowledge my supervisor Prof G.C.E Mbah for supervising me though I wasn't assigned to him, but he accepted me wholeheartedly and was like a father to me. I also appreciate his patience, time, encouragement, and supervision. I enjoyed every bit of it. Thank you so much, sir.

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And to my wonderful siblings, Kelechi, Onyi, and Favour for their show of love and encouragement, and lastly, to my parents who always encouraged, and supported me both financially and otherwise. God bless you all.

ABSTRACT

Family is the bedrock of any society. This is formed as a result of marriage. In this work, we described marriage and divorce, factors that influence or cause divorce, its effects, and possible remedies. When conflicts arise there are ways of resolving the conflicts/problems, but when this fails and the couple feels they are no longer compatible it results in divorce. We formulated and theoretically analyzed the mathematical model of divorce in marriage with five compartmental equations. We analyzed the equilibrium points of the model and investigated their stability at these points. The local stability of the DFE and EEP was established using the Jacobian matrix.

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CHAPTER ONE

1.0 INTRODUCTION:

Marriage and divorce are two important social institutions that have a great impact on the individual, families, and society at large.

Marriage is the formal and socially recognized union of two people to become husband and wife. Divorce is the dissolution or end of a marital relationship. Family is the vital cell of any society, hence it makes up the society and this is formed as a result of marriage. People get married for the following reasons; love, childbearing, companionship, legal status, financial security, to create a safe relationship where one can reach the deepest level of intimacy and connection, to provide security for children, to signify a life-long commitment and based on ones' religious belief.

It is presumed that divorce was not part of the original plan after getting married. This is because marriage is supposed to be a union for life without any thought that it may one day come to an end. But life happens, things change, love fades, problem arises and excitement begins to wave off. Divorce does not just happen. It is a result of accumulated, unresolved, unspoken errors, unchecked attitudes, and overlooked misunderstandings that sponsor divorce in the long run.

It is paramount for couples to understand that marriage is not a bed of roses and would not be smooth always, enables a marriage to have high hopes of surviving. There is no perfect marriage, every marriage has its problems or challenges peculiar to it. But, there are some factors associated with a successful marriage such as developing communication skills, commitment,

forgiveness, accepting each other, appreciation, spending quality time, respect, and trust for each other. This is because everyone desires peace of mind and not a nagging husband or wife.

The rate of divorce has fluctuated over time and can depend on individual characteristics like age at marriage, education level, income, gender, religious belief, personality traits, relationship factors like communication, conflict resolution skills and presence of children, social factors like social norms, expectations surrounding marriage, availability of social support and economic conditions. Some processes in divorce involve preparing legal paperwork, going to court, dividing assets/debts, child support, custody, or visitation. Divorce has both positive and negative effects on children, couples, and society.

Mathematical modeling is a versatile tool that can be used to solve a wide range of problems by identifying the problem, developing a mathematical representation of the problem, the validity of the model, and using the model to make predictions and decisions. Mathematical modeling allows us to formulate formal representations of marriage and divorce dynamics, capturing the intricate interactions between individuals, their choices, and external influences. By employing mathematical techniques and computational simulations, researchers can explore and quantify the impacts of various factors on marriage and divorce rates, predict future trends, and evaluate the effectiveness of potential interventions.

1.1 AIM OF STUDY →

The aim of the study is to develop a mathematical model of marriage divorce that can be used to understand the factors that influence divorce in marriage.

1.2 OBJECTIVES OF THE STUDY

- To review all existing and relevant literature on the mathematical model of marriage divorce.
- Discuss the factors that influence the rate of divorce in marriage
- Discuss the effects of divorce and the effectiveness of different intervention strategies in reducing divorce rates and promoting stable marriages.
- To develop a mathematical model that can predict the likelihood of divorce in marriage.

1.3 SCOPE OF THE STUDY

- To understand the dynamics of marriage and divorce.
- To draw a mathematical model of divorce in marriage.
- To analyze the stability of the model.
- To determine the models by comparing their parameters to real-world data.

1.4 SIGNIFICANCE OF THE STUDY

- To develop a model that captures the complex interaction and factors influencing divorce in marriage, by understanding the dynamics and underlying mechanisms of marriage and divorce.
- From my study, future marriage and divorce can be predicted and these can enable researchers and policy makers to understand the importance of societal changes on marriage and divorce rates.

- My study shows the relationship between mathematics and other fields of discipline like social sciences, and education by applying mathematical modeling techniques to understand complex social phenomena. This is achieved by the use of quantitative methods to represent human relationships. This interdisciplinary approach can inspire further collaborations between researchers in different fields and foster innovation in both mathematical modeling and social sciences.

1.5 LIMITATIONS OF THE STUDY

- The model involves simplifications and assumptions. These simplifications may not capture the full complexity of real-world relationships, and the model's assumptions may not hold universally or in all cultural contexts.
- The data used for validating the model may be limited in scope, geographic area, or period. Also, the accuracy and reliability of the data sources can vary.
- Due to the complexity of marriage and divorce, there is a limitation in capturing all relevant variables and interactions by the model. There might be uncertainty in the prediction and projections made by the model.

CHAPTER TWO

2.0 LITERATURE REVIEW

In this chapter, we review some research work on a Mathematical Model of Divorce in Marriage.

Ogwumu et al. (2020) worked on a "mathematical model for predicting the rate of divorce Tendencies in Nigeria". However, they did theoretical work on the topic by describing factors that influence divorce and how an increase in these factors or a decrease in these factors affects marriage and can lead to divorce. A sampling technique was carried out and questionnaires retrieved from those who were not single, widows, and widowers were considered. The factors considered were; if the age difference (A) is low, there are zero or no changes in the divorce rate and if the age difference is high, there is a high chance for divorce. Hence, divorce is directly proportional to their age differences i.e. $D \propto A$. The lower the educational difference (E) of married couples, the lesser or nil chances for divorce to exist, and the higher the educational differences, the higher the chances for divorce i.e. $D \propto E$. They consider the case where the high earning is from the man or woman. If she earns more than the husband (high income difference), there is a tendency to higher divorce chances otherwise, there is less or no divorce i.e. $D \propto I_D$. If the total income of the family is high/normal, there will be minimal chances of divorce in the marriage and there will be a likelihood of divorce if the income of the couple is lower than their demands. Hence, divorce is inversely proportional to their income i.e. $D \propto \frac{1}{I_T}$. They thought that no matter how good a marriage is, it is not free from all of the factors promoting divorce in marriage. The DFE, EEP, and stability were not calculated, rather the author used linear optimization, least square, and correlation analysis to validate their model.

Tessema et al. (2022) worked on "Mathematical model Analysis of Marriage Divorce" and considered divorce as a transmitted disease that transmits between humans, propagated by divorced men/women over married ones. They considered four subpopulations: those who have reached the age of getting married which they called "single individuals denoted by $S(t)$ ", those who are married denoted by " $M(t)$ ", those who separate but not divorced; called the "Broken marriage denoted by $B(t)$ ", and those who are divorced denoted by $D(t)$. Hence, they have four equations. They also considered two forms of death rate namely; the Natural death rate (μ), and the Death rate of individuals due to divorce (σ). The invariant region, the divorce-free, and endemic equilibrium points, local and global stability analysis were calculated. The model bifurcation analysis was done and exhibits forward bifurcation at $R_0 = 1$. In our work, the single individuals were not considered, because divorce can only exist in a married setting. When conflicts set in, we have three cases that were considered. In case one, they can be counseled by friends/family, pastors, or marriage counselors. Secondly, they can separate, and lastly eventually divorce.

Gambrab and Adzadu, (2018) worked on the "Mathematical model of divorce epidemic in Ghana". They proposed and analyzed a non-linear MSD (married, separation, and divorce) model to study the dynamics of the divorce epidemic in Ghana. They assumed that sex, race, and social status do not affect the probability of being divorced and that Divorce occurs in a closed environment. They were of the opinion that reducing the contact rate between married and divorced, increasing the number of marriages going into separation, and educating the separators to refrain from divorce can be useful in combating the divorce epidemic. They discussed the existence and stability of divorce-free and endemic equilibrium and performed the sensitivity analysis of the reproduction numbers.

Duato and Jódar (2013) worked on "Mathematical modeling of the spread of divorce in Spain". They viewed the spread of divorce to be an economic and social contagion that is propagated by personal contact between divorced and married women. They were of the opinion that divorce rules should be equitable for both sexes and should not favor either women or men. This is because most divorce rules favor women and a proportion of women abuse this as a measure of claiming divorce. Also; their opinion was that a suitable measure to reduce the growth in the number of divorces can be derived by focusing on girls' education, even from childhood periods, teaching them how to fill their leisure time effectively. When the divorce rate is reduced it is only important for social stability and the health of children but also maintain the birth rate and pensions in the population. They generated an SMD (Single, Married, Divorce) model. They used an age category from 20 to 60 years to generate statistics from the years 2000 to 2009 where they considered singles who got married from 2000 to 2009 and those who got divorced within 2000- 2009. In this work, only three classes were considered, single, married, and, divorced. There was no room for separation or counseling. In our work, the single class was not considered because divorce occurs among married people, those who develop problems in their marriage can go for counseling, separate, or eventually divorce.

Tessema et al. (2022) worked on "Analysis of the Mitigation Strategies for Marriage Divorce from Mathematical Modeling Prospective". They were of the opinion that ways of fighting divorce are couple relationship educators, reducing getting married too young, and consulting separators to renew their marriage. They formulated a mathematical model and extended it into an optimal control model. The model was in four compartments SMBD (Single, Married, Broken, and Divorced class). With the assumption that sex, race, and social status do not affect the probability of getting divorced. An optimal control theory was used to establish conditions

under which divorce in marriage can be stopped and to examine the impact of a possible combination of these controls on reducing divorce in the population. Three control intervention was given namely; couple relationship education (u_1), reducing getting married too young (u_2), and consulting separators to renew their marriage (u_3). These three control interventions were incorporated into the model equations.

Lhous et al. (2017) worked on "Discrete mathematical modeling and optimal control of the marital status: the monogamous marriage case". In this work, an optimal control is investigated to reduce the number of virgin, and divorced populations and to increase the number of married ones. The first control introduced the benefits of an awareness campaign to educate virgin men and women about the benefits of marriage for individuals or society, and the second control characterized the legal procedures, administrative complications, and the heavy financial and social consequences of divorces. A discrete version of Pontryagin's maximum principle is used to analyze the optimal control problem, and a numerical simulation is given to illustrate the obtained result. They classified individuals in the society into eight compartments namely, virgin men, virgin women, married men, married women, divorced men, divorced women, widowed men, and widowed women. Hence, they developed eight equations for the VMDW model.

CHAPTER THREE

3.1 Marriage:

The simplest definition of marriage is the union between two people (man and woman) to become husband and wife. English Dictionary defines "marriage as a union of two or more people". It is the sole means to enter a family and plays the role of primarily providing for the needs and requirements of its members. Marriage according to Havilland (2000) is "a relationship between one or more men and one or more women recognized by society as having a continuing claim to the right of sexual access to one another. Marriage is a transaction and resulting contract in which a woman and a man are recognized by society as having a continuing claim to the right of sexual access to one another, and in which the woman involved is eligible to bear children (Havilland 1996). It is also a union between a man and a woman such that the children born to the woman are recognized as legitimate offspring of both partners.

There are three major forms of marriage namely; Monogamy, Polygamy, and group marriage. Monogamy is the universal type of marriage that exists in every society and this form of marriage involves one man and one woman. Monogamy is divided into two types; straight monogamy which gives no room for remarriage when one's spouse dies or is divorced and serial monogamy where one can remarry after divorce or death of one's spouse.

Polygamy is another popular form of marriage. This involves a marriage between a man and more than one woman (Polygyny) or between one woman and more than one man (Polyandry). Polygyny is of two forms; Sororal (where a man marries two sisters at once. It is usually observed in places where they pay high bride price) and Non-Sororal (where a man marries more than one woman but his wives are not blood-related). Polyandry is of two types; Fraternity

(Here, two brothers take a single woman as their wife. This is usually practiced by tribes of India called the Todas tribe) and Non-Fraternity (here, multiple men take up one woman as their wife. But the men are not blood-related).

Group marriage involves a group of men who are married to a group of women and they all have a common husband and wives. There is also same-sex marriage also called gay marriage (marriage of people with the same legal sex). But, our concentration is on monogamy or polygyny form of marriage.

3.2 Divorce

The Cambridge Dictionary defines divorce as the official or legal process to end a marriage. It is the dissolution of marriage by a court or any competent body. This is also a delicate accident in a marital relationship. It is also a permanent separation of couples as a result of unexpected marriage outcomes. When problems arise and all techniques put in place to resolve these problems like counseling from friends, family, and priests, separation fails then divorce is the last option.

3.3 Separation

This shouldn't be confused with divorce. Here, the couple is legally married but lives apart. The couples cannot remarry unless their separation has been finalized. Separation does not have any financial implications; you will still be considered married and will be responsible for your spouse's debts, unlike divorce. When problems arise and they fail to reconcile, they tend to separate and go their separate ways and remain unmarried because of their children.

3.4 How can Marriage be sustained

Most people enter into marriage with so much expectation but it is important to have an understanding that every marriage has problems/challenges that are peculiar to it. Hence there is no perfect marriage anywhere. There will be mood swings, stress, and boredom during this period of marriage.

Marriage is sacred and a lifetime commitment commandment by God. When a couple takes marital vows "in sickness and in health, in riches, and in poverty entails that marriage is not a bed of roses, so it wouldn't be rosy to the end. Having an understanding that "marriage takes work" and once work has been performed, the reward is a happy and successful home. It involves both parties and teamwork to achieve this. Hence, below are some ways in which marriage can be sustained:

Communication:

Having effective and clear communication is the key to any relationship. Healthy communication leads to more trust among couples making it easier to deal with conflicts, and unnecessary arguments and build a stronger partnership. This enables one to tell how he/she feels and be open to his/her spouse without any reservations. Communication involves verbal and non-verbal exchanges between the two people involved. Lack of communication can lead to blame, relationship anxiety, depression, and resentment in marriage and there is a high risk of breaking, separation, or divorce. Some important communication techniques include; listening, thinking before you speak, participating with sincerity and honesty (i.e. no silent treatment), and observing your partner's body language is an important non-verbal communication technique.

Respect:

This is an important pillar of a strong and lasting marriage. When couples respect each other, they are likely to communicate effectively, resolve conflict peacefully, and support each other through good and bad times. There should be mutual respect both in public and in private. Respect is reflected in how you treat each other daily (i.e. your ability to respect each other opinion). Respect does not imply controlling someone or making them do what you want them to do, it is the ability/freedom to be their selves and be loved for who they are. They are ways of showing respect to your spouse like listening to your spouse with an open mind, considering your spouse's feelings (before saying something, think about how it can affect your partner), appreciating your spouse's contribution, never taking each other for granted and always remember that your spouse is a valuable and irreplaceable part of your life.

Love:

This is a powerful force in any relationship but love is not enough; hence cannot function alone, it needs other factors like communication, commitment, respect, and acceptance. This is the foundation of a strong marriage and it is important to show your love in action not just words by doing things like giving them a nice treat, cooking for them, massaging or even running errands for them, and being supportive. Love provides room for forgiveness, and makes marriage fun, interesting/exciting because when you are in love, he or she is happy and fulfilled.

Commitment:

This is essential for a successful marriage. It implies the willingness of two people to stay both in good times and bad to work through problems by putting the needs of their spouse first. It also implies their willingness to communicate honestly and openly and their willingness to compromise. When you commit to your marriage, you are creating a foundation that helps you withstand any storm.

Making time for each other as a couple:

Couple time/ alone time is very important in a marriage. Making out time out of your busy schedule shows how much you value the person and the relationship. These are moments that hold much value spent in the company of those who are dear to you. You can spend quality time with your spouse by going on a vacation, taking a walk in a park, or enjoying a coffee break together, it can also be staying indoors and watching movies or spending time before bedtime. It helps in strengthening connections, forming deeper bonds, and building memories together. When couples fail to spend quality time together, they feel you are too busy for them and might begin to fall out of love and end the relationship.

Trust:

This is another ingredient to sustain a marriage. It implies believing that your spouse is reliable and honest and has your interest at heart. One can build trust by being open, honest, reliable, dependable, and transparent with your spouse in terms of your feelings, thoughts, words,

3.6 Reasons for Divorce in Marriage

It is presumed that when a couple gets married, divorce is not part of the initial plan. Divorce cannot happen in a vacuum hence, there cannot be divorce without marriage. People opt for divorce for reasons best known to them. Some reasons are;

Lack of Commitment:

This is a major factor that contributes to divorce. It can be seen as the unwillingness to work on the relationship and putting little or no effort into the marriage by developing a nonchalant attitude towards issues/challenges facing them. To ensure that the marriage is successful, both parties have a role to play and must be intentional/willing and committed to each other this can lead to distancing, resentment, and divorce eventually. Partners who are not committed become emotionally unavailable (no longer provide the love, support, and attention they need), unwilling to compromise (giving away something that would make the relationship not work), become unfaithful (infidelity is a major sign of lack of commitment. Here, one partner cheats on the other and this can be a devastating betrayal that can lead to divorce)

Domestic violence:

This can also be called domestic abuse or intimate partner violence. This is a pattern of behavior in any relationship that is used to gain or maintain power or control over an intimate partner. It can be physical, emotional, psychological, or sexual actions that affect the other partner and this makes it difficult to make a happy marriage. When one's partner is abusive, it makes it difficult for the other partner to feel safe, and secure in the relationship and this leads to a breakdown in communication and trust. It can also lead to fear and can influence divorce because he or she feels that's the only way out of such violence.

Issues of intimacy:

Intimacy is an important aspect of any healthy relationship. When couples deny their spouses' sex or any form of intimacy and see themselves as roommates or flatmates instead of partners or when one partner has a higher or lower libido than the other, it breeds conflicts in marriage. This can lead to a feeling of loneliness, rejection, and being sex-starved, and one might decide to find satisfaction elsewhere which now results in cheating if they are unable to meet each other's needs. If this cannot be controlled, then it may lead to divorce.

Unforgiveness:

Unforgiveness is the inability or unwillingness to forgive your spouse for upsetting, betraying, or breaking your trust. Unforgiveness brings about anger and resentment in the marriage. It makes it difficult to communicate effectively because it can lead to arguments, fights, and even

violence, unresolved problems in marriage and there is a feeling of distance/detachment because the pain of your spouse's offense is still there. Unforgiveness makes it difficult for people to move on from their past and this leads to a lack of trust and makes it difficult to build a new future together.

Incompatibility:

This refers to the lack of alignment between two people in their values, goals, beliefs, or behavior. It can be a major source of divorce conflict and tension. When two people have different goals, it can be difficult to stay on the same page and work towards a common goal. Incompatibility makes it difficult for people to build a shared life together, plan for the future, and raise children.

Communication problems:

Problems like lack of commitment, trust issues, unforgiveness, issues of intimacy, incompatibility, etc. can be resolved by effective communication. But when there is a communication breakdown, couples cannot express their thoughts or feelings to each other without arguments, fights, and misunderstandings playing out.

Dr. John Gottman and his wife, a clinical psychologist Dr. Julie Gottman developed a model called "*The Gottman model*" which identifies four behavioral traits or factors that contribute to marital breakdown (divorce). They are;

Criticism:

This is the first sign that a marriage is headed for divorce. There is a difference between complaint and criticism. The first tends to focus on one specific behavior or action while the latter attacks an individual's entire character and leaves the person feeling worthless. It involves always finding faults and pointing out your spouse's flaws.

Contempt:

This is the biggest predictor of divorce. This is an act of belittling, name-calling, eye-rolling, hostility, sarcasm, expressing disgust, and making your partner feel inferior. Contempt is harmful in every way- physically, emotionally, and psychologically.

Defensiveness:

This is a response to criticism and contempt. It is referred to as a counter-attack, lashing out, making excuses, and blaming them for your problems. The reason why it is toxic is that people are not taking responsibility for their problems.

Stonewalling:

Here, a partner builds a wall to block the other partner out. It involves a withdrawal from communication, refusing to talk to a partner, listening to them, and pretending to be occupied.

3.7 Effects of Divorce

Divorce comes with a lot of trauma and stress. Its effects on individuals, children, and society cannot be over-emphasized.

3.7.1 Effects on Children

Academic problem:

It can result in poor performance in their academics. The more they are distracted, they can't focus or concentrate on their school work. It can lead to an increase in absenteeism and if not attended to can result in them being school dropouts.

Relationship problem:

Children of divorced parents may have difficulty forming and maintaining stable relationships. They suffer social isolation; struggle to interact with others and feel insecure

Emotional distress:

They become emotionally destabilized, and feelings of anger, confusion, loss, anxiety, feeling of guilt, signs of depression can appear, and many more.

Behavioral problem:

Children of divorced parents are likely to exhibit behavioral problems like aggression, use of illicit substances, and teenage pregnancy when divorced parents struggle to make time for their children, hence they are easily influenced by their peer groups.

3.7.2 Effects on the Individuals

- Change in physical health: Spouses may experience changes in their physical health such as weight loss or gain, headaches and struggle to maintain a healthy life. They can suffer from heart attacks, strokes, cancer, and/or a weakened immune system.

- Emotional distress like feelings of guilt, stress, depression, insomnia, confusion/sadness, sadness, and anxiety.
- They might experience financial problems as they may have to adjust to living on a single income and paying alimony/child support.
- There is a change in social life. They may lose contact with family and friends who were connected to the spouse. Many struggle to give love a second try because of past experiences with their previous spouses. If they eventually find love, they struggle with trust; hence they can't maintain a healthy relationship.

3.7.3 Effect on the society

The effect of divorce in society is classified into positive and negative aspects.

- Positive effects:

Increased productivity:

Individuals who are no longer tied down to unhappy/unhealthy relationships are more likely to focus on their careers and pursue new opportunities and their goals/interests which can lead to an increase in economic productivity.

Increased social mobility:

This is the opportunity to move to new places and start a new life and this can lead to improved social mobility.

Increased social equality:

This helps to break down traditional gender roles and give women more economic and social power. This allows women to be more independent and self-sufficient; hence they are no longer dependent on their spouses and are more likely to pursue their goals and careers.

- Negative effects:

Increased poverty:

It breeds poverty, particularly for women and children in the society. This is because women are more likely to be the primary caregivers for children after a divorce and may experience difficulty in finding and maintaining employment.

Increased crime:

This can result in an increase in crime among young people. Young people who are exposed to divorce are likely to experience emotional distress and behavioral problems which can lead to their risk of engaging in criminal activities.

Decreased social cohesion:

This results in weakening the bonds between family members and friends by disrupting social networks and making it difficult for people to maintain relationships.

Increased social instability:

This results in social instability which makes it difficult for the family to function effectively and creates uncertainty about the family.

Increased emotional and behavioral problems in their children.

3.8 Remedies/Solutions to Divorce

- Counseling: Couples should go for counseling at least a couple of times per year, but most people wait till the problem is beyond their control before seeking help.
- Communication is key
- Spending quality time together
- Accepting each other's differences
- Understanding the issues in one's marriage enables one to seek solutions
- Be accountable to each other
- Starting over from scratch. It is important to remember why you both fell in love, got married, and build your marriage all over again
- Remove divorce as an option in your marriage.

3.9 THE MODEL:

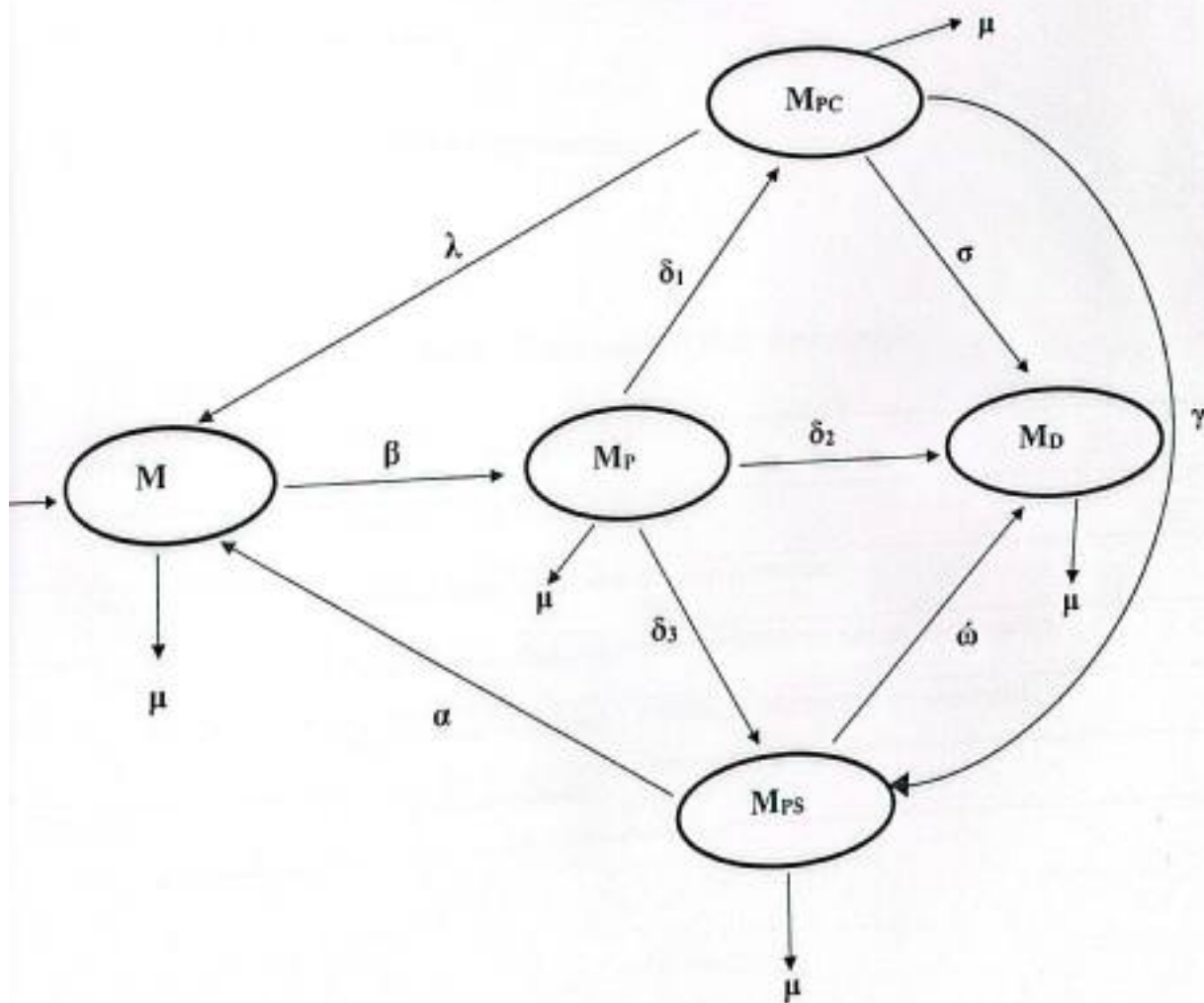
3.9.1 Assumptions of the model:

- Divorce can occur in a marriage setting
- Every married couple can develop problems.
- Every married couple who develop problem can either go for counseling, separation, or divorce.
- Every married couple who develops problems and goes for counseling can separate, divorce, or reunite.

- Every married couple who develops problems and is separated, can either reunite or divorced.

3.9.2 Flow Chart:

Based on the assumptions, the flow chart showing the process of marriage and divorce is shown below:



3.9.3 Model Equations

$$\frac{dM}{dt} = n - \beta M - \mu M + \alpha M_{PS} + \lambda M_{PC} \quad (1)$$

$$\frac{dM_P}{dt} = \beta M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P \quad (2)$$

$$\frac{dM_{PC}}{dt} = \delta_1 M_P - \sigma M_{PC} - \lambda M_{PC} - \gamma M_{PC} - \mu M_{PC} \quad (3)$$

$$\frac{dM_{PS}}{dt} = \gamma M_{PC} + \delta_3 M_P - \alpha M_{PS} - \omega M_{PS} - \mu M_{PS} \quad (4)$$

$$\frac{dM_D}{dt} = \delta_2 M_P + \sigma M_{PC} + \omega M_{PS} - \mu M_D \quad (5)$$

Where $\beta = a_1 M_D + a_2 M_{PS}$, a_1 and a_2 are constants

Table 1. Model Variables and their Descriptions

Variables	Descriptions
M	Fully married individuals
M _P	Married individuals who develop problems
M _{PC}	Married individuals who develop problems and go to counseling
M _{PS}	Married individuals who develop problems and are separated
M _D	Divorced individuals

Table 2: Model Parameters and Their Description

Parameters	Description
n	Newly married couples
β	The rate at which married individuals develop problems in their marriage
ω	The rate at which married individuals develop problems and are separated eventually results in divorce
μ	Natural death rate of individuals
λ	The rate at which married individuals develop problems, go for counseling, and eventually reunite
σ	The rate at which married individuals who develop problems, go for counseling and eventually divorce
α	The rate at which married individuals who develop problems and separate eventually reunite
γ	The rate at which married individuals who develop problems, go for counseling and eventually separate
δ_1	The rate at which married people who develop problems go for counseling
δ_2	The rate at which married people who develop problems eventually divorce
δ_3	The rate at which married people who develop problems separate

3.9.4 Description of the model:

From the first compartment, the newly married couple join to make up the married individuals. The married individuals can be reduced by natural death. Those who experience problems in their marriage and go for counseling or are separated can also reunite to make up the married population.

From the second compartment, the rate of married people with problems have three options; they can go for counseling, separate or divorce. The second compartment can also be reduced by natural death (μ).

From the third compartment, those who develop problems in their marriage and go for counseling can be reduced by natural death, they can separate, divorce or reunite to make up the married people.

From the fourth compartment, those who develop problems in their marriage and are separated can be reduced by natural death, can unite back to been married, or divorce. Those that who experience problem and go for counseling can also increase the separation class.

From the last compartment, those who develop problems in marriage, those who go for counseling, and those who are separated can make up the divorce class. This compartment can only be reduced by natural death (μ).

CHAPTER FOUR

4.0 ANALYSIS OF THE MODEL:

To verify if the model is biologically meaningful and well posed, we determine the invariant region to break that all solutions of our model equations are positive for all time $(t) > 0$ and that Ω is indeed, bounded.

4.1 INVARIANT REGION:

This region can be obtained by this theorem;

We claim that the solution of system (1-5) is feasible for $t > 0$ if they enter the invariant region.

Summing all the five equations of the model gives;

$$N(t) = M(t) + M_P(t) + M_{PC}(t) + M_{PS}(t) + M_D(t)$$

Then, differentiating $N(t)$ with respect to t ,

$$\frac{dN}{dt} = \frac{dM}{dt} + \frac{dM_P}{dt} + \frac{dM_{PC}}{dt} + \frac{dM_{PS}}{dt} + \frac{dM_D}{dt}$$

$$\frac{dN}{dt} = (n - \beta M - \mu M + \alpha M_{PS} + \lambda M_{PC}) + (\beta M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P) + (\delta_1 M_P - \sigma M_{PC} - \lambda M_{PC} - \gamma M_{PC} - \mu M_{PC}) + (\gamma M_{PC} + \delta_3 M_P - \alpha M_{PS} - \omega M_{PS} - \mu M_{PS}) + (\delta_2 M_P + \sigma M_{PC} + \omega M_{PS} - \mu M_D)$$

$$\frac{dN}{dt} = n - \mu M - \mu M_P - M_{PC} - \mu M_{PS} - \mu M_D$$

$$\frac{dN}{dt} = n - \mu N$$

$$\text{Where } N = M + M_P + M_{PC} + M_{PS} + M_D$$

$$\frac{dN}{dt} + \mu N = n$$

(4.1.1)

The equation 4.1.1 is a linear ODE. The integrating factor is given as $q^{\int \mu dt} = q^{\mu t}$

Multiplying both sides of equation 4.1.2 By I.F = $q^{\mu t}$, we have

$$q^{\mu t} \left(\frac{dN}{dt} + \mu N \right) = n q^{\mu t}$$

$$(q^{\mu t} N(t))' = n q^{\mu t}$$

Integrating both sides of the inequality.

$$N(t) q^{\mu t} = \frac{n}{\mu} q^{\mu t} + C_1 N(t) = C_1 q^{-\mu t} + \frac{n}{\mu} \quad (4.1.2)$$

Using the initial condition $t=0$

$$N(0) = C_1 q^{-\mu t}$$

$$\text{Then } C_1 = N(0) - \frac{n}{\mu}$$

This equation 4.1.2 becomes

$$N(t) = (N(0) - \frac{n}{\mu}) q^{-\mu t} + \frac{n}{\mu}$$

$$N(t) = \frac{n}{\mu} [1 - q^{-\mu t}] + N(0) q^{-\mu t}$$

$$\text{as } t \rightarrow \infty, \text{ Then, } N(t) \leq \frac{n}{\mu}$$

Hence, the total population size $N(t) \rightarrow \frac{n}{\mu}$ as $t \rightarrow \infty$. Therefore all the feasible solution set of the population of the model enters the region $N = \{(M, M_P, M_{PC}, M_{PS}, M_D) \in R_+^5 \mid M(t) \geq 0, M_P \geq 0, M_{PC}(t) \geq 0, M_{PS} \geq 0, M_D \geq 0, N(t) \leq \frac{n}{\mu}\}$

In conclusion, the model is well-defined and the region is positively invariant for all t .

4.2 POSITIVITY OF SOLUTION:

Here, the model equation remains positive i.e.

Let the initial data $\{M(0), M_P(0), M_{PC}(0), M_{PS}(0), M_D(0)\}$

Proof:

From the first equation in the model we have,

$$\frac{dM}{dt} = n - \beta M - \mu M + \alpha M_{PS} + \lambda M_{PC}$$

$$\frac{dM}{dt} \geq -\beta M - \mu M$$

By method of separation of variables, we have

$$\frac{dM}{dt} \geq -(\beta + \mu) dt$$

Integrating both sides

$$\ln M \geq -(\beta + \mu)t + C_1$$

$$M \geq e^{-(\beta + \mu)t + C_1}$$

$$M(t) \geq A e^{-(\beta + \mu)t}, \quad \text{where } A = e^{C_1}$$

$$\text{at } t = 0, M(0) \geq A$$

$$M(t) \geq M(0) e^{-(\beta+\mu)t}$$

Thus, $M(t) \geq 0$ Since $(\beta + \mu) > 0$ and $\forall t \geq 0$

From equation (2),

$$\frac{dM_P}{dt} = \beta M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P$$

$$\frac{dM_P}{dt} \geq -\mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P$$

$$\frac{dM_P}{dt} \geq -(\mu + \delta_1 + \delta_2 + \delta_3)M_P$$

By method of Separation of variables, we have

$$\frac{dM_P}{M_P} \geq -(\mu + \delta_1 + \delta_2 + \delta_3)dt$$

Integrating both sides,

$$\ln(M_P) \geq -(\mu + \delta_1 + \delta_2 + \delta_3)t + C_2$$

$$M_P(t) \geq e^{-(\mu + \delta_1 + \delta_2 + \delta_3)t + C_2}$$

$$M_P(t) \geq B e^{-(\mu + \delta_1 + \delta_2 + \delta_3)t}, \text{ where } e^{C_2} = B$$

$$M_P(0) \geq B, \text{ at } t = 0$$

$$M_P(t) \geq M_P(0) e^{-(\mu + \delta_1 + \delta_2 + \delta_3)t}$$

$$\text{Thus, } M_P(t) \geq 0, \forall t \geq 0$$

Similarly, from equation (3)

$$\frac{dM_{PC}}{dt} = \delta_1 M_P - \sigma M_{PC} - \lambda M_{PC} - \gamma M_{PC} - \mu M_{PC}$$

$$\frac{dM_{PC}}{dt} \geq -\sigma M_{PC} - \lambda M_{PC} - \gamma M_{PC} - \mu M_{PC}$$

$$\frac{dM_{PC}}{dt} \geq -(\sigma + \lambda + \gamma + \mu)M_{PC}$$

By method of Separation of variables, we have

$$\frac{dM_{PC}}{M_{PC}} \geq -(\sigma + \lambda + \gamma + \mu)dt$$

Integrating both sides,

$$\ln(M_{PC}) \geq -(\sigma + \lambda + \gamma + \mu)t + C_3$$

$$M_{PC}(t) \geq e^{-(\sigma + \lambda + \gamma + \mu)t + C_3}$$

$$M_{PC}(t) \geq C e^{-(\sigma + \lambda + \gamma + \mu)t}, \text{ where } e^{C_3} = C$$

$$M_{PC}(0) \geq C \text{ at } t=0$$

$$M_P(t) \geq M_{PC}(0) e^{-(\sigma + \lambda + \gamma + \mu)t}$$

$$\text{Thus, } M_{PC}(t) \geq 0 \cdot \forall t \geq 0$$

From the fourth equation,

$$\frac{dM_{PS}}{dt} = \gamma M_{PC} + \delta_3 M_P - \alpha M_{PS} - \omega M_{PS} - \mu M_{PS}$$

$$\frac{dM_{PS}}{dt} \geq -(\alpha M + \omega M + \mu)M_{PS}$$

By method of Separation of variables, we have

$$\frac{dM_{PS}}{M_{PS}} \geq -(\alpha M + \omega M + \mu)dt$$

Integrating both sides,

$$\ln(M_{PS}) \geq -(\alpha M + \omega M + \mu)t + C_4$$

$$M_{PS}(t) \geq e^{-(\alpha M + \omega M + \mu)t + C_4}$$

$$M_{PS}(t) \geq D e^{-(\alpha M + \omega M + \mu)t}$$

$$M_{PS}(t) \geq D e^{-(\alpha M + \omega M + \mu)t}$$

$$M_{PS}(0) \geq D \text{ at } t=0$$

$$M_{PS}(t) \geq M_P(0) e^{-kt}$$

$$\text{Thus, } M_{PS}(t) \geq 0, \forall t \geq 0$$

From the fifth equation,

$$\frac{dM_D}{dt} = \delta_2 M_P + \sigma M_{PC} + \omega M_{PS} - \mu M_D$$

$$\frac{dM_D}{dt} \geq -\mu M_D$$

By method of Separation of variables, we have

$$\frac{dM_D}{M_D} \geq -\mu dt$$

Integrating both sides,

$$\ln(M_D) \geq -\mu t + C_4$$

$$M_D(t) \geq e^{-\mu t} + e^{C_4}$$

$$M_D(t) \geq E e^{-\mu t}, \text{ where } e^{C_4} = E$$

$$M_D(0) \geq E, \text{ at } t=0$$

$$M_D(t) \geq M_D(0) e^{-\mu t}$$

$$\text{Thus, } M_D(t) \geq 0, \forall t \geq 0$$

The above proofs show that the model equations of the system are all positive for all time because $e^u > 0$, for all $u \in \mathbb{R}$

4.3 EQUILIBRIUM STATE

There are two equilibriums evaluated here, the disease-free equilibrium (DFE) and the endemic equilibrium point (EEP).

4.3.1 DISEASE-FREE EQUILIBRIUM (DFE)

This is a state where there is no conflict in the marriage. It is denoted by

$$E^0 = (M^0, M_P^0, M_{PC}^0, M_{PS}^0, M_D^0) \text{ where } M^0 \neq 0, M_P^0 = 0, M_{PC}^0 = 0, M_{PS}^0 = 0, M_D^0 = 0$$

$$\text{At equilibrium, } \frac{dM}{dt} = \frac{dM_P}{dt} = \frac{dM_{PC}}{dt} = \frac{dM_{PS}}{dt} = \frac{dM_D}{dt} = 0$$

From equation (1),

$$\frac{dM}{dt} = n - \beta M^0 - \mu M^0 + \alpha M_{PS}^0 + \lambda M_{PC}^0$$

$$0 = n - (\beta + \mu) M^0, \text{ where } \beta = a_1 M_D + a_2 M_{PS}$$

$$\mu M^0 = n$$

$$M^0 = \frac{n}{\mu}$$

$$\text{Therefore at (DFE), } E^0 = (M^0, M_P^0, M_{PC}^0, M_{PS}^0, M_D^0) = \left(\frac{n}{\mu}, 0, 0, 0, 0\right)$$

4.3.2 ENDEMIC EQUILIBRIUM POINT (EEP)

The endemic equilibrium point is the point where the conflicts cannot be eradicated but remain in the population. In the presence of conflicts, then $M^* \neq 0, M_P^* \neq 0, M_{PC}^* \neq 0, M_D^* \neq 0$. The model has an equilibrium point called the endemic equilibrium denoted by E^* and is given as $E^* = (M^*, M_P^*, M_{PC}^*, M_{PS}^*, M_D^*)$, where $(M^*, M_P^*, M_{PC}^*, M_{PS}^*, M_D^*) > 0$

To derive the EEP, we solve the model equations by equating it to zero.

$$0 = n - \beta M - \mu M + \alpha M_{PS} + \lambda M_{PC} \quad (1)$$

$$0 = \beta M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P \quad (2)$$

$$0 = \delta_1 M_P - \sigma M_{PC} - \lambda M_{PC} - \gamma M_{PC} - \mu M_{PC} \quad (3)$$

$$0 = \gamma M_{PC} + \delta_3 M_P - \alpha M_{PS} - \omega M_{PS} - \mu M_{PS} \quad (4)$$

$$0 = \delta_2 M_P + \sigma M_{PC} + \omega M_{PS} - \mu M_D \quad (5)$$

From equation 2,

$$0 = \beta M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P$$

$$0 = \beta M^* - \mu M_P^* - \delta_1 M_P^* - \delta_2 M_P^* - \delta_3 M_P^*$$

$$\beta M^* = \mu M_P^* + \delta_1 M_P^* + \delta_2 M_P^* + \delta_3 M_P^*$$

$$\beta M^* = (\mu + \delta_1 + \delta_2 + \delta_3) M_P^*$$

$$M^* = \frac{(\mu + \delta_1 + \delta_2 + \delta_3) M_P^*}{\beta}$$

$$M^* = \frac{A M_P^*}{\beta}, \text{ Where } (\mu + \delta_1 + \delta_2 + \delta_3) = A$$

4.3.2.1

From equation 3,

$$0 = \delta_1 M_P - \sigma M_{PC} - \lambda M_{PC} - \gamma M_{PC} - \mu M_{PC}$$

$$0 = \delta_1 M_P^* - \sigma M_{PC}^* - \lambda M_{PC}^* - \gamma M_{PC}^* - \mu M_{PC}^*$$

$$\delta_1 M_P^* = \sigma M_{PC}^* + \lambda M_{PC}^* + \gamma M_{PC}^* + \mu M_{PC}^*$$

$$\delta_1 M_P^* = (\sigma + \lambda + \gamma + \mu) M_{PC}^*$$

$$M_{PC}^* = \frac{\delta_1 M_P^*}{\sigma + \lambda + \gamma + \mu}$$

$$M_{PC}^* = \frac{\delta_1 M_P^*}{B} \text{ Where, } \sigma + \lambda + \gamma + \mu = B$$

4.3.2.2

From equation 4,

$$0 = \gamma M_{PC} + \delta_3 M_P - \alpha M_{PS} - \omega M_{PS} - \mu M_{PS}$$

$$0 = \gamma M_{PC}^* + \delta_3 M_P^* - \alpha M_{PS}^* - \dot{\omega} M_{PS}^* - \mu M_{PS}^*$$

$$\delta_3 M_P^* + \gamma M_{PC}^* = \alpha M_{PS}^* + \dot{\omega} M_{PS}^* + \mu M_{PS}^*$$

$$M_{PS}^* = \frac{\delta_3 M_P^* + \gamma M_{PC}^*}{\alpha + \dot{\omega} + \mu}$$

$$\text{but, } M_{PC}^* = \frac{\delta_1 M_P^*}{B}$$

$$M_{PS}^* = \frac{\delta_3 M_P^* + \gamma \frac{\delta_1 M_P^*}{B}}{\alpha + \dot{\omega} + \mu} \quad \text{let, } \alpha + \dot{\omega} + \mu = C$$

$$M_{PS}^* = \frac{\delta_3 M_P^* + \gamma \frac{\delta_1 M_P^*}{B}}{C}$$

$$M_{PS}^* = \frac{M_P^* (\delta_3 + \frac{\delta_1 \gamma}{B})}{C}$$

$$M_{PS}^* = \frac{M_P^* (B\delta_3 + \delta_1 \gamma)}{CB}$$

4.3.2.3

From equation 5,

$$0 = \delta_2 M_P + \sigma M_{PC} + \dot{\omega} M_{PS} - \mu M_D$$

$$0 = \delta_2 M_P^* + \sigma M_{PC}^* + \dot{\omega} M_{PS}^* - \mu M_D^*$$

$$0 = \delta_2 M_P^* + \sigma M_{PC}^* + \dot{\omega} M_{PS}^* - \mu M_D^*$$

$$\mu M_D^* = \delta_2 M_P^* + \sigma M_{PC}^* + \dot{\omega} M_{PS}^*$$

$$\text{but, } M_{PC}^* = \frac{\delta_1 M_P^*}{B}, \quad M_{PS}^* = \frac{M_P^* (B\delta_3 + \delta_1 \gamma)}{CB}$$

$$\mu M_D^* = \delta_2 M_P^* + \frac{\delta_1 \sigma M_P^*}{B} + \frac{\dot{\omega} M_P^* (B\delta_3 + \delta_1 \gamma)}{CB}$$

$$\mu M_D^* = M_P^* \left(\delta_2 + \frac{\delta_1 \sigma}{B} + \frac{\omega(B\delta_3 + \delta_1 \gamma)}{CB} \right)$$

$$M_D^* = \frac{1}{\mu} \left[M_P^* \left(\delta_2 + \frac{\delta_1 \sigma}{B} + \frac{\omega(B\delta_3 + \delta_1 \gamma)}{CB} \right) \right]$$

4.3.2.4

Therefore, $E^* = (M^*, M_P^*, M_{PC}^*, M_{PS}^*, M_D^*)$

$$E^* = \left(\frac{AM_P^*}{\beta}, M_P^*, \frac{\delta_1 M_P^*}{B}, \frac{M_P^* (B\delta_3 + \delta_1 \gamma)}{CB}, \frac{1}{\mu} M_P^* \left(\delta_2 + \frac{\delta_1 \sigma}{B} + \frac{\omega(B\delta_3 + \delta_1 \gamma)}{CB} \right) \right)$$

4.4 STABILITY:

4.4.1 Local Stability of Disease-Free Equilibrium

To analyze the local stability of the disease-free equilibrium, we obtain the Jacobian matrix of our equations at the disease-free equilibrium point.

The system of equation (1- 5) is stable if the eigenvalues of the Jacobian transformation evaluated at Disease Free Equilibrium are negative and asymptotically stable if the system is stable and the Jacobian $\det. J > 0$ and $\text{tr. } J < 0$.

By Jacobian transformation,

Let,

$$\frac{dM}{dt} = n - (\beta - \mu)M + \alpha M_{PS} + \lambda M_{PC} = n - (a_1 M_D + a_2 M_{PS} + \mu)M + \alpha M_{PS} + \lambda M_{PC} = A$$

$$\frac{dM_P}{dt} = \beta M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P = (a_1 M_D + a_2 M_{PS})M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P = B$$

$$\frac{dM_{PC}}{dt} = \delta_1 M_P - \sigma M_{PC} - \lambda M_{PC} - \gamma M_{PC} - \mu M_{PC} = C$$

$$\frac{dM_{PS}}{dt} = \gamma M_{PC} + \delta_3 M_P - \alpha M_{PS} - \omega M_{PS} - \mu M_{PS} = D$$

$$\frac{dM_D}{dt} = \delta_2 M_P + \sigma M_{PC} + \omega M_{PS} - \mu M_D = E$$

Where, $\beta = a_1 M_D + a_2 M_{PS}$, and a, b are constants.

$$J^0 = \begin{bmatrix} \frac{\partial A}{\partial M} & \frac{\partial A}{\partial M_P} & \frac{\partial A}{\partial M_{PC}} & \frac{\partial A}{\partial M_{PS}} & \frac{\partial A}{\partial M_D} \\ \frac{\partial B}{\partial M} & \frac{\partial B}{\partial M_P} & \frac{\partial B}{\partial M_{PC}} & \frac{\partial B}{\partial M_{PS}} & \frac{\partial B}{\partial M_D} \\ \frac{\partial C}{\partial M} & \frac{\partial C}{\partial M_P} & \frac{\partial C}{\partial M_{PC}} & \frac{\partial C}{\partial M_{PS}} & \frac{\partial C}{\partial M_D} \\ \frac{\partial D}{\partial M} & \frac{\partial D}{\partial M_P} & \frac{\partial D}{\partial M_{PC}} & \frac{\partial D}{\partial M_{PS}} & \frac{\partial D}{\partial M_D} \\ \frac{\partial E}{\partial M} & \frac{\partial E}{\partial M_P} & \frac{\partial E}{\partial M_{PC}} & \frac{\partial E}{\partial M_{PS}} & \frac{\partial E}{\partial M_D} \end{bmatrix}$$

$$J^0 = \begin{bmatrix} -(\beta + \mu) & 0 & \lambda & -aM + \alpha & -aM \\ \beta & -(\mu + \delta_1 + \delta_2 + \delta_3) & 0 & aM & aM \\ 0 & \delta_1 & -(\sigma + \lambda + \gamma + \mu) & 0 & 0 \\ 0 & \delta_3 & \gamma & -(\alpha + \omega + \mu) & 0 \\ 0 & \delta_2 & \sigma & \omega & -\mu \end{bmatrix}$$

Evaluated at DFE, $E^0 = (M^0, M_P^0, M_{PC}^0, M_{PS}^0, M_D^0) = (\frac{n}{\mu}, 0, 0, 0, 0)$

$$J^0 = \begin{bmatrix} -\mu & 0 & \lambda & -\frac{a_2 n}{\mu} + \alpha & -\frac{a_1 n}{\mu} \\ 0 & -\mu - \delta_1 & 0 & \frac{a_2 n}{\mu} & \frac{a_1 n}{\mu} \\ 0 & \delta_1 & -\lambda - \mu & 0 & 0 \\ 0 & 0 & 0 & -\alpha - \mu & 0 \\ 0 & 0 & 0 & 0 & -\mu \end{bmatrix}$$

Find the Eigenvalues, $|J^0 - hI| = 0$

Where h = Eigenvalues,

$$J^0 = \begin{vmatrix} -\mu - h & 0 & \lambda & -\frac{a_2 n}{\mu} + \alpha & -\frac{a_1 n}{\mu} \\ 0 & -\mu - \delta_1 - h & 0 & \frac{a_2 n}{\mu} & \frac{a_1 n}{\mu} \\ 0 & \delta_1 & -\lambda - \mu - h & 0 & 0 \\ 0 & 0 & 0 & -\alpha - \mu - h & 0 \\ 0 & 0 & 0 & 0 & -\mu - h \end{vmatrix} = 0$$

$$(-\mu - h)(-\mu - \delta_1 - h)(-\lambda - \mu - h)(-\alpha - \mu - h)(-\mu - h) = 0$$

$$h_1 = -\mu \text{ (twice)}, h_2 = -(\mu + \delta_1), h_3 = -(\lambda + \mu), h_4 = -(\alpha + \mu)$$

Since all the eigenvalues are negative (that is all $h < 0$), $\text{tr. } J^0 < 0$ and $\det. J^0 > 0$ then the system is locally and asymptotically stable.

4.42 Local Stability at Endemic Equilibrium Point

The system of equation (1-5) is locally stable at Endemic Equilibrium if $R_0 > 0$ or the eigenvalues are negative.

Similarly,

By Jacobian transformation,

Let,

$$\frac{dM}{dt} = n - (\beta - \mu)M + \alpha M_{PS} + \lambda M_{PC} = n - (a_1 M_D + a_2 M_{PS} + \mu)M + \alpha M_{PS} + \lambda M_{PC} = A$$

$$\frac{dM_P}{dt} = \beta M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P = (a_1 M_D + a_2 M_{PS})M - \mu M_P - \delta_1 M_P - \delta_2 M_P - \delta_3 M_P = B$$

$$\frac{dM_{PC}}{dt} = \delta_1 M_P - \sigma M_{PC} - \lambda M_{PC} - \gamma M_{PC} - \mu M_{PC} = C$$

$$\frac{dM_{PS}}{dt} = \gamma M_{PC} + \delta_3 M_P - \alpha M_{PS} - \omega M_{PS} - \mu M_{PS} = D$$

$$\frac{dM_D}{dt} = \delta_2 M_P + \sigma M_{PC} + \omega M_{PS} - \mu M_D = E$$

Where, $\beta = a_1 M_D + a_2 M_{PS}$, and a, b are constants.

$$J = \begin{bmatrix} \frac{\partial A}{\partial M} & \frac{\partial A}{\partial M_P} & \frac{\partial A}{\partial M_{PC}} & \frac{\partial A}{\partial M_{PS}} & \frac{\partial A}{\partial M_D} \\ \frac{\partial B}{\partial M} & \frac{\partial B}{\partial M_P} & \frac{\partial B}{\partial M_{PC}} & \frac{\partial B}{\partial M_{PS}} & \frac{\partial B}{\partial M_D} \\ \frac{\partial C}{\partial M} & \frac{\partial C}{\partial M_P} & \frac{\partial C}{\partial M_{PC}} & \frac{\partial C}{\partial M_{PS}} & \frac{\partial C}{\partial M_D} \\ \frac{\partial D}{\partial M} & \frac{\partial D}{\partial M_P} & \frac{\partial D}{\partial M_{PC}} & \frac{\partial D}{\partial M_{PS}} & \frac{\partial D}{\partial M_D} \\ \frac{\partial E}{\partial M} & \frac{\partial E}{\partial M_P} & \frac{\partial E}{\partial M_{PC}} & \frac{\partial E}{\partial M_{PS}} & \frac{\partial E}{\partial M_D} \end{bmatrix}$$

$$J^* = \begin{bmatrix} -(\beta + \mu) & 0 & \lambda & \frac{-a_2 M}{\mu} + \alpha & \frac{-a_1 M}{\mu} \\ \beta & -(\mu + \delta_1 + \delta_2 + \delta_3) & 0 & \frac{a_2 M}{\mu} & \frac{\mu}{a_1 M} \\ 0 & \delta_1 & -(\sigma + \lambda + \gamma + \mu) & \frac{\mu}{0} & \frac{\mu}{0} \\ 0 & \delta_3 & \gamma & -(\alpha + \omega + \mu) & 0 \\ 0 & \delta_2 & \sigma & \omega & -\mu \end{bmatrix}$$

Evaluated at EEP,

$$E^* = \left(\frac{AM_P^*}{\beta}, M_P^*, \frac{\delta_1 M_P^*}{\beta}, \frac{M_P^*(B\delta_3 + \delta_1 \gamma)}{CB}, \frac{1}{\mu} M_P^* \left(\delta_2 + \frac{\delta_3 \sigma}{\beta} + \frac{\omega(B\delta_3 + \delta_1 \gamma)}{CB} \right) \right)$$

Set,

$$\frac{M_P^*(B\delta_3 + \delta_1 \gamma)}{CB} = M_P^* \left(\delta_2 + \frac{\delta_3 \sigma}{\beta} + \frac{\omega(B\delta_3 + \delta_1 \gamma)}{CB} \right) = P, (\mu + \delta_1 + \delta_2 + \delta_3) = S, (\sigma + \lambda + \gamma + \mu) = T$$

and $(\alpha + \omega + \mu) = V$

$$J^* = \begin{bmatrix} -(a_1 P + a_2 M + \mu) & 0 & 0 & -\frac{a_2 AM_P^*}{\mu} & -\frac{a_1 AM_P^*}{\mu} \\ a_1 P + a_2 M & -S & 0 & \frac{a_2 AM_P^*}{\mu} & \frac{a_1 AM_P^*}{\mu} \\ 0 & \delta_1 & -T & \frac{\mu}{0} & \frac{\mu}{0} \\ 0 & \delta_3 & \gamma & -V & 0 \\ 0 & \delta_2 & \sigma & \omega & -\mu \end{bmatrix}$$

Adding the first row with the second row gives,

$$J^* = \begin{bmatrix} -\mu & -S & 0 & 0 & 0 \\ a_1 P + a_2 M & -S & 0 & \frac{a_2 AM_P^*}{\mu} & \frac{a_1 AM_P^*}{\mu} \\ 0 & \delta_1 & -T & \frac{\mu}{0} & \frac{\mu}{0} \\ 0 & \delta_3 & \gamma & -V & 0 \\ 0 & \delta_2 & \sigma & \omega & -\mu \end{bmatrix}$$

Finding the Eigen values, $|J^* - hI| = 0$, where $h = \text{Eigen value}$

$$J^* = \begin{bmatrix} -\mu - h & -S & 0 & 0 & 0 \\ a_1 P + a_2 M & -S - h & 0 & \frac{a_2 A M_E^*}{\omega} & \frac{a_1 A M_E^*}{\omega} \\ 0 & \delta_1 & -T - h & \mu & \mu \\ 0 & \delta_3 & \gamma & -V - h & 0 \\ 0 & \delta_2 & \sigma & \omega & -\mu - h \end{bmatrix}$$

$$(-\mu - h)(-S - h)(-T - h)(-V - h)(-\mu - h) = 0$$

$$h_1 = -\mu(\text{twice}), h_2 = -S, h_3 = -T, h_4 = -V < 0$$

Since all the eigenvalues are negative and the $\text{tr. } J^* < 0$ and $\text{det. } J^* > 0$, we conclude that the endemic equilibrium is locally and asymptotically stable.

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